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Research Paper

Nusselt number and friction factor correlations for laminar flow in parallelogram serpentine micro heat exchangers

Tong-Miin Liou*, Chun-Sheng Wang, Hsun Wang

Department of Power Mechanical Engineering, National Tsing Hua University, Hsinchu 30013, Taiwan

HIGHLIGHTS

- Heat transfer in parallelogram serpentine micro heat exchangers is firstly explored.
- Effects of aspect ratio (α), included angle (θ), and Reynolds number (*Re*) are studied.
- A new finding is that there exists a critical value of α for Nusselt number (*Nu*).
- A critical *Re* is found for Fanning friction factor (f) as well.
- *Nu* and f correlations are newly proposed as composite functions of α, θ, and *Re*.

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ABSTRACT

The topic of compact micro heat exchanger is one of the most active areas in heat and mass transfer research today. However, there is a lack of literatures on serpentine channels with parallelogram cross-section. This article provides numerical study of the hydraulic and thermal performance in parallelogram serpentine channels with aspect ratios (α) ranging from 0.25 to 4 and wall included angles (θ) ranging from 45° to 90°. The Reynolds number (*Re*), characterized by channel hydraulic diameter and mean bulk velocity, is varied from 100 to 300. Their influences on the fluid flow and heat transfer characteristics are investigated under constant wall temperature condition. It is newly found that Nusselt number (*Nu*) distributions on bottom and top walls in slant wall channels ($\theta \neq 90°$) are asymmetric and there exists a critical value of α and *Re* for *Nu* and Fanning friction factor (f), respectively. Furthermore, two new correlations for overall *Nu* and f with respect to α, θ, and *Re* are proposed for laminar flow in parallelogram channels, with maximum errors less than 10% and 20%, respectively, compared with numerical results.

1. Introduction

In recent decades, compact heat exchangers have been one of the most interesting research areas due to the rapid growth of power consumption and miniaturization of electromechanical systems. To optimize the heat exchanger performance in a limited dimension, there are many types of tortuous channel patterns that have been developed. One of the most common ones is the serpentine channel, which is widely applied to various fields such as underfloor heating/cooling systems, turbine blade/vane internal cooling, on-chip cooling, solar thermal collector, and proton exchange membrane fuel cells (PEMFC). PEMFC is one of the most promising power sources because of higher efficiency (> 40%), shorter start-up time, lower temperature range (50–100 °C), and cleaner emissions (water only). Therefore, increasing studies about PEMFC applications have been reported in various fields such as transportation industries, stationary power systems, and portable electronic devices. To ensure long durability and high efficiency of PEMFC, proper thermal management techniques are indispensable since PEMFC generates the amount of the waste heat comparable to its power output and only tolerates a small temperature deviation from its design point [1]. According to Faghri and Guo [2], most fuel cell units below 2 kW can be cooled with increasing reactant air, but the ones beyond 2 kW usually need a separated cooling channel.

Chen et al. [3] conducted a numerical analysis to study the laminar fluid flow and heat transfer between solid plates in coupled cooling process. The uniformity of temperature across the cooling plates was evaluated for three types of serpentine channels and three types of parallel channels. The prediction showed that the cooling effect of serpentine channels could be better than that of parallel ones. Inoue et al. [4] numerically explored the optimal separator shape and flow

* Corresponding author.
E-mail address: tmliou@pme.nthu.edu.tw (T.-M. Liou).

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Nomenclature

a	channel width, mm
b	channel height, mm
c_p	specific heat, J/kg K
D_h	hydraulic diameter, mm ($4ab\sin\theta/(2a + 2b)$)
f	Fanning friction factor ($2\tau_w/\rho U_b^2$)
h	convective heat transfer coefficient, W/m^2 K
k	thermal conductivity of working fluids, W/m K
L	channel length, mm
Nu	local Nusselt Number (hD_h/k)
P	pressure, Pa
Pr	Prandtl number ($\nu/(k/\rho c_p)$)
q'	wall heat transfer rate per unit axial length, W/m
q''	heat flux, W/m^2
Re	Reynolds number ($U_b D_h/\nu$)
s	normalized axial distance
T	temperature, K
T^*	normalized temperature ($(T-T_w)/(T_w-T_{inlet})$)
TPF	thermal performance factor ($(Nu_m/Nu_0)/(f/f_0)^{1/3}$)
$\boldsymbol{U}$	velocity vector, m/s ($u\boldsymbol{i} + v\boldsymbol{j} + w\boldsymbol{k}$)
u^*	normalized x-velocity (u/U_b)
v^*	normalized y-velocity (v/U_b)
w^*	normalized z-velocity (w/U_b)
w_d	divider thickness, mm ($(a + b)/2$)
x	x-coordinate, mm
x^*	normalized x-coordinate ($2x/(a + b)$)
y	y-coordinate, mm
y^*	normalized y-coordinate (y/b)
z	z-coordinate, mm
z^*	normalized z-coordinate in the first ($(z + 0.5W_d)/a$) and second pass ($(z-0.5W_d)/a$)
z^{**}	normalized z-coordinate in the turn ($z/(a + 0.5W_d)$)

Greek symbols

α	channel aspect ratio (a/b)
θ	included angle, degree
ν	kinematic viscosity, m^2/s
ρ	fluid density, kg/m^3
τ	shear stress, Pa

Subscripts

b	bulk average
m	axially-weighted average
s	peripherally-averaged axially local
w	wall
0	fully developed flow in circular channel

pattern of gas and cooling water for fuel cell durability. The examined separator types included an ordinary serpentine separator and a distributed serpentine separator. It was found that the ordinary serpentine separator provides the most uniform relative humidity distribution. A new channel patterns design, termed convection-enhanced serpentine flow-field, was presented by Xu and Zhao [5]. This design was shown to enhance the convective heat transfer under bipolar plate ribs. Yu et al. [6] published their research on six multi-pass serpentine channels in cooling plates based on computational fluid dynamics (CFD) simulations. The results demonstrated that multi-pass serpentine channels lead to better cooling performance compared with a conventional serpentine channel, in terms of both the maximum temperature and temperature uniformity. Kurnia et al. [7] extended CFD simulations to more cooling channel designs including parallel, serpentine, wavy, coiled and novel hybrid channels. It was concluded that the coiled channel shows the best heat transfer as well as uniformity of temperature, but the pressure drop penalty is relatively high. According to the aforementioned studies, cooling plates with serpentine channels generally exhibit better performance in terms of heat transfer and temperature uniformity compared with those with parallel channels. However, serpentine channels not only augment the heat transfer but also increase the pressure drop due to the turn-induced secondary flow. Therefore, a thorough understanding of hydrothermal behaviors inside serpentine channels is essential to the design of heat exchanger.

Compared to straight channels, tortuous channels significantly enhance the heat transfer performance due to their turn-induced secondary flow. Therefore, many researchers have devoted themselves to figuring out how the channel pattern influences the flow and heat transfer behaviors. In 1927, Dean [8] firstly reported the formation of the secondary flows (Dean vortices) in turn configuration through experiment. Dennis [9] conducted a numerical solution of Navier-Stokes equations for steady laminar flow in a slightly curved tube with constant diameter. He found that the formation of the vortices is dependent on a dimensionless parameter named Dean Number (De), which is the product of the Reynolds number (Re, based on the axial mean flow velocity and the hydraulic diameter) and the square root of the curvature ratio. To address the effect of the curvature ratio, Wang and Liu [10] numerically analyzed a forced convection flow in slightly curved (curvature ratio = 5×10^{-6}) microchannel with square cross-section. It was found that, no matter how small the channel curvature ratio is, there always exists a turn-induced secondary flow in the cross-plane which increases the friction factor moderately and the Nusselt number significantly. Wang and Chyu [11] numerically investigated the secondary flow patterns and the heat transfer distributions in a two-pass square duct with three different turn configurations, i.e. straight-corner turn, rounded-corner turn, and circular turn. They summarized that the straight-corner configuration demonstrates the highest turn-induced heat transfer augmentation whereas the circular turn one shows the weakest. Chintada et al. [12] conducted numerical simulations of laminar flow and heat transfer in square serpentine channels with right-angle turns. The flow and heat transfer fields were solved with different geometry parameters and different Re for air and water. They concluded that the heat transfer performances of serpentine channels are better and worse than that of straight channels for water and air respectively. Rogusati et al. [13] carried out a numerical study of the heat transfer distributions in a serpentine tube with various wavelength and bend's curvature for fully developed laminar flow. It was found that the Dean vortices in serpentine channel get stronger after each bend if the bend curvature is in the same direction as the previous one. Furthermore, Dean vortices are found to inhibit flow separation and gradually dominate the flow filed as Re becomes higher. At higher values of Re (corresponding to $De > 150$), a second pair of vortices develops, and the flow becomes unsteady for $Re > 200$. They also argued that the alignment of the flow with the vorticity in these structures allows strong mixing of the fluid without creating large pressure drops. Additionally, the heat transfer enhancement due to Dean vortices gets larger at higher values of Prandtl number (Pr), but with less increment in pressure drop. This is consistent with what Chintada et al. [12] has reported. Lasbet et al. [14] performed numerical simulations of fluid flow and heat transfer in three-dimensional c-shaped channels for $Re \approx 200$. With the novel design, the flow field inside the channel generated chaotic advection which preserves the high thermal performance in a limited size of the bipolar plates. However, the manufacturing cost and the thickness of the cooling plate could also be increased because of the three-dimensional channel geometry. For tortuous channels at higher Re, fluid particle trajectories become chaotic and the turn-induced secondary flow starts stretching and folding. This phenomenon, named chaotic advection, is firstly proposed by Aref [15], and it dominates the thermal hydraulic features in tortuous channels. Zheng et al. [16] conducted numerical studies for steady-state laminar flows and heat transfer in periodic zigzag channels with a square cross-section. The computational domain consisted of up to 14 repeating zigzag units with smoothly joined inlet and outlet sections. The results showed that fully-developed periodic flow fields can be achieved for $Re < 200$. However, a spatially periodic flow field and heat transfer distribution disappears for higher Re due to the enhanced effect of chaotic advection and the flow field becomes transient for $Re > 400$. They extended this work to the semi-circular and zig-zag channels [17,18], and further concluded that the thermal hydraulic behavior is very sensitive to the inlet conditions and a proportional relation is found between the mean Nusselt number (Nu) and $Pr^{1/3}$. Except for the afore-mentioned channel patterns, there also exists many literatures on different channel patterns such as trapezoid [19,20] and wavy [21,22]. Nonetheless, each of them could be viewed as a special case of serpentine channels.

Previous studies primarily focus on the tortuous channels with circular or square cross-sections. However, in practical applications, the cross-sectional shape of cooling channels may not be ideally circular or rectangular because of manufacturing facts or space limitations. Earlier studies about laminar forced convection flow in straight ducts with different cross-sectional shapes are compiled by Shah and London [23]. They concluded that the cross-sectional shape of the channel affects the heat transfer performance greatly. Rogusati et al. [24] numerically investigated the laminar flow and heat transfer behavior in serpentine channels with semi-circular cross-section. The serpentine patterns were characterized by their wavelength, channel diameter, and curvature radius of bend for $Re < 450$, and for a range of geometric configurations for $Re = 110$. It was found that the flow field is characterized by the Dean vortices in each bend. As the Re increases, more complex vorticial flow patterns emerge and the flow domain is increasingly dominated by these vortices. Also, they concluded that the alignment of flow with vorticity leads to efficient fluid mixing and high rates of heat transfer, and this is also consistent with their previous study [13].

Some applications of cooling channels not only focus on overall heat transfer performance but also demand asymmetric heat load or even one-side insulation. For example, one side of cooling plate in PEMFC is subjected to almost 90% waste heat generated in cathode [25], the leading and trailing walls of turbine vanes/blades face unequal heat loads, and the non-target side of underfloor heating/cooling systems is expected to be insulated [26]. Therefore, the asymmetric heat transfer behavior of parallelogram channel is a promising feature for these applications. Liou et al. [27] performed PIV measurements and flow visualization in a two-pass smooth parallelogram channel with 45° included angle for $Re = 10{,}000$. It was found that the secondary flow patterns are strongly asymmetric, especially in and after the bend region, as opposed to those of square channels [28]. They further extended their studies with infrared thermometry measurements on a rotating model for $Re = 5000$–$20{,}000$ [29] and found that the single side heat transfer performance of parallelogram channel is superior to that of square channels. More recently, another numerical study of rotating two-pass smooth rhombic channel with included angles ranging from 45° to 135° was also presented by Liou et al. [30]. The flow and heat transfer fields were solved with low-Re realizable k-ε model and enhanced wall treatment at $Re = 10{,}000$. They concluded that the included angle has a significant influence on the hydrothermal characteristics in both stationary and rotating channels, and the overall heat transfer performance of the channel with 60° included angle is better than that of other cases examined as the rotation number is higher than 0.2. Although these turbulent fluid flow and heat transfer investigations report many interesting results, few works have been devoted to the exploration of laminar flows in parallelogram serpentine channels.

In summary, many interesting results indicating the effects of cross-sectional shape, channel pattern, and turn configuration have been reported. However, to the best of authors' knowledge, most of the studies in the open literature do not simultaneously examine the effect of the aspect ratio (α) and the included angle (θ) of the parallelogram serpentine channels on the asymmetric flow and heat transfer behaviors. Hence, the aim of the present work is to explore the combined effects of α and θ on the fluid flow and heat transfer features in parallelogram serpentine channels. The paper also evaluates overall thermal performance based on the constant pumping power consumption. To accomplish these objectives, a finite volume method is adopted to simultaneously simulate developing laminar flows in parallelogram channels with various θ ranging from 45° to 90° and α ranging from 0.25 to 4 under constant wall temperature condition which is recommended by Liou et al. [31]. The findings of this study could be applied to various fields, such as fuel cells (Fig. 1), combustion turbine engines, solar thermal collectors, and underfloor heating/cooling systems which have a demand for unequal heat load on opposite sides.

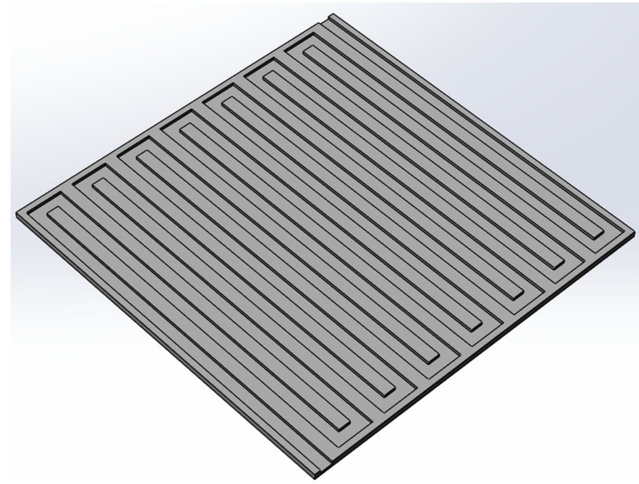

Fig. 1. Schematic of serpentine micro channel with square cross-section for PEMFC.

2. Description of problem

Fig. 2(a) and (b) demonstrate a schematic graph of the parallelogram channel with different α and θ, and the corresponding computational mesh, respectively. For brevity, only a unit (two passes) of the serpentine channel is considered in the present study. The two walls ($y^* = 0.5$ and $y^* = -0.5$) normal to y-direction are denoted as top wall and bottom wall, respectively, and the others are named as inner wall and outer wall. For convenience, the x-coordinates downstream and upstream of the turn are respectively negative and positive by specifying the coordinate origin at the center of divider tip. Correspondingly, the longitudinal mean velocity component u takes plus sign following the main flow.

3. Computational details

3.1. Numerical methods

The following assumptions are made for the problem presented above:

(1) steady incompressible laminar flow,
(2) constant fluid properties (water, $Pr = 7$),
(3) neglecting viscous dissipation, gravitation, and radiation.

Then, the corresponding governing equations become

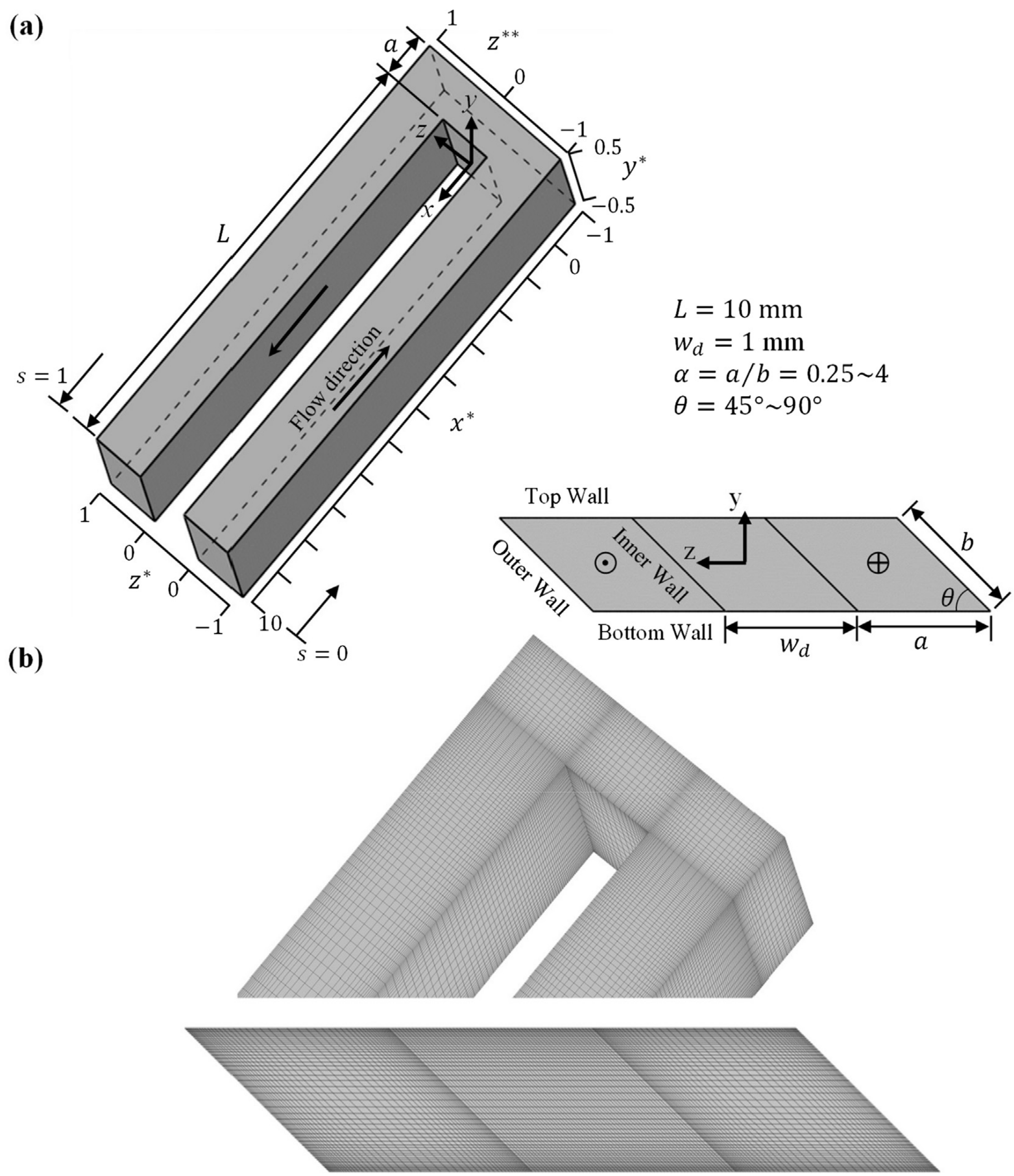

Fig. 2. (a) 3-D computational geometry and coordinate and (b) the corresponding enlarged view of numerical grids for $\alpha = 1$ and $\theta = 45°$.

Continuity equation:

$$\nabla\cdot\boldsymbol{U} = 0 \tag{1}$$

Momentum equation:

$$\boldsymbol{U}\cdot\nabla\boldsymbol{U} = \frac{-1}{\rho}\nabla P + \nu\nabla^2\boldsymbol{U} \tag{2}$$

Energy equation:

$$\boldsymbol{U}\cdot\nabla T = \frac{k}{\rho c_p}\nabla^2 T \tag{3}$$

In this study, the commercial software package, FLUENT 16.0 [32], based on cell-centered finite-volume method is employed to solve the continuity equation, Navier–Stokes equations, and energy equation. SIMPLE algorithm is adopted for velocity–pressure coupling with the algebraic multigrid method while the second-order upwind scheme is utilized for spatial discretization in pressure term. The momentum and energy equations are solved with a QUICK scheme which is typically more accurate on structured meshes aligned with the flow direction [33]. Least squares cell-based gradient evaluation is applied for predicting scalar values at the cell faces and for computing secondary diffusion terms and velocity derivatives. To reduce the artificial numerical diffusion and save the computational resources, computational domains are meshed by hexahedral elements with refinement near the wall as shown in Fig. 2(b). To reduce iterative errors, the convergence of simulations is declared as all global scaling residuals are less than 10^{-6} according to the guidance of Fluent 16.0 [32] and previous literatures [12,34].

3.2. Boundary conditions

A uniform velocity U_b and temperature T_{inlet} are specified at the inlet of the channel while no-slip condition is imposed on all the walls. A constant wall temperature T_w is applied to the walls. The distributions of velocity and temperature are assumed to be with zero-gradient in the direction normal to the exit plane while the outlet pressure profile is linearly extrapolated. The mathematical forms of the boundary conditions are given below.

$$\boldsymbol{U}_{inlet} = (U_b \quad 0 \quad 0),\ T_{inlet} = 283\,\mathrm{K} \tag{4}$$

$$\boldsymbol{U}_{w} = (0 \quad 0 \quad 0),\ T_{w} = 303\,\mathrm{K} \tag{5}$$

$$\left.\frac{\partial \boldsymbol{U}}{\partial n}\right|_{outlet} = (0 \quad 0 \quad 0),\ \left.\frac{\partial \boldsymbol{T}}{\partial n}\right|_{outlet} = 0 \tag{6}$$

3.3. Data reduction

To investigate the effects of cross-sectional configurations on the explored channel flow and heat transfer behaviors, the aspect ratio (α) and the included angle (θ) vary from 0.25 to 4 and 45° to 90°, respectively (Fig. 2(a) and Table 1). In the present work, the Reynolds number, characterized by channel hydraulic diameter and working fluid of water, is ranged from 100 to 300

$$Re = \frac{U_b D_h}{\nu} \tag{7}$$

where U_b is the bulk velocity in the ducts and D_h is the hydraulic diameter expressed below.

$$D_h = \frac{2ab\sin\theta}{a+b} \tag{8}$$

To simultaneously maintain the same Re and volume flow rate, all channels have the same perimeter of 4 mm. and the channel inlet U_b in the present work varies from case to case due to the different*ineachcase* in each case as listed in Table 1.

To evaluate the pressure drop in each case, the Fanning friction factor (f) is defined as

$$f = -\frac{1}{4}\frac{\Delta P}{L}\frac{2D_h}{\rho U_b^2} = \frac{2\tau_{w,s}}{\rho U_b^2} \tag{9}$$

where ΔP and L respectively denote the pressure drop across the channel and the length of the channel; *denotesperipherally−averagedwallshearstress*. denotes peripherally-averaged wall shear stress.

$$\tau_{w,s} = \frac{1}{P_s}\int_{P_s}\tau_w \cdot dl \tag{10}$$

To assess the heat transfer performance, the local, peripherally-averaged, and mean heat transfer coefficients are defined as

$$h = \frac{q''}{T_w - T_b} \tag{11}$$

$$h_s = \frac{q'}{T_{w,s} - T_b} \tag{12}$$

$$h_m = \int h_s ds \tag{13}$$

where T_w is the local wall temperature. T_b represents the bulk temperature at a given normalized axial location s and is given by

$$T_b = \frac{1}{\dot{m}c_p}\int_{A_s}\rho c_p T\boldsymbol{U}\cdot d\boldsymbol{A} \tag{14}$$

q' is the wall heat transfer rate per unit axial length defined as

Table 2
Grid independence test.

Grid size	#grid	fRe	diff. (%)	Nu_m	diff. (%)
10 × 10 × 130	13,000	22.56	2.90	8.70	1.44
20 × 20 × 260	104,000	23.24	0.69	8.83	1.04
30 × 30 × 390	351,000	23.40	0.46	8.92	0.53
40 × 40 × 520	832,000	23.51	–	8.97	–

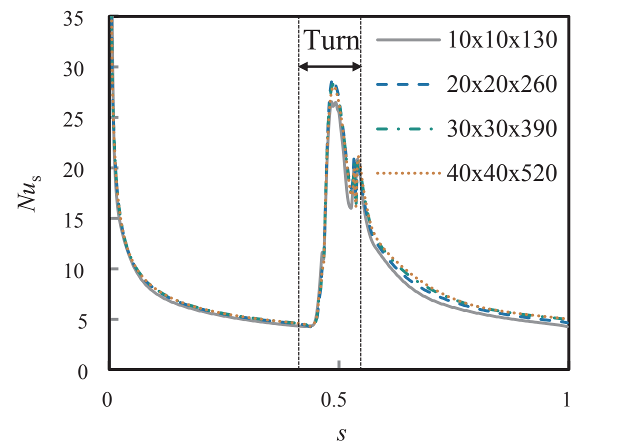

Fig. 3. Grid independence test under constant wall temperature condition ($\alpha = 1$, $\theta = 45°$, and $Re = 300$).

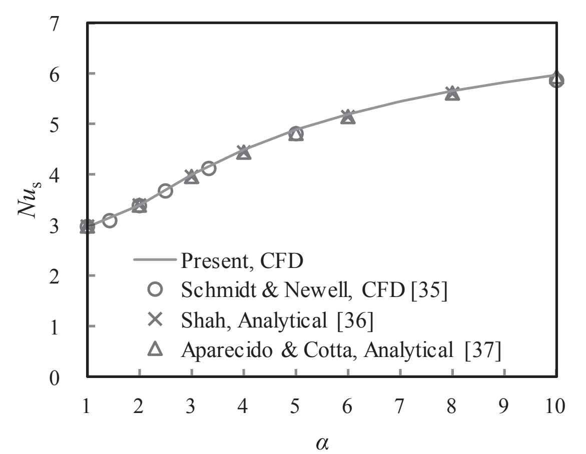

Fig. 4. Comparisons between presently predicted Nusselt number (Nu_s) and previous results for fully developed straight rectangular duct flows with various α.

$$q' = \frac{1}{P_s}\int_{P_s} q''\cdot dl \tag{15}$$

$T_{w,s}$ defined by Eq. (16) denotes the peripherally-averaged wall temperature at a given normalized axial location s.

Table 1
Dimensions of the microchannels studied.

α	a (μm)	b (μm)	Hydraulic diameter, D_h (μm)									
			θ = 45°	50°	55°	60°	65°	70°	75°	80°	85°	90°
0.25	400	1600	453	490	524	554	580	601	618	630	638	640
0.5	670	1330	630	683	730	772	808	837	861	878	888	891
1	1000	1000	707	766	819	866	906	940	966	985	996	1000
2	1330	670	630	683	730	772	808	837	861	878	888	891
4	1600	400	453	490	524	554	580	601	618	630	638	640

$$T_{w,s} = \frac{1}{P_s}\int_{P_s} T_w \cdot dl \tag{16}$$

Then, the Nusselt number is defined as:

$$Nu = \frac{hD_h}{k} \tag{17}$$

Note that Nu can be evaluated at any position on the wall, at any given axial location, or as a globally-averaged value with h, h_s, and h_m, respectively.

Based on the constant pumping power consumption, the overall thermal performance factor (TPF) for each case is evaluated as

$$TPF = \frac{Nu_m/Nu_0}{(f/f_0)^{1/3}} \tag{18}$$

3.4. Computational grid

To test grid independence as well as to save the computational resources, several grid sizes are studied for the case of $\theta = 45^{\circ}$ with $\alpha = 1$ and $Re = 300$ as tabulated in Table 2. From the grid size $30 \times 30 \times 390$ to the grid size $40 \times 40 \times 520$, the refinement of grid results in improvements of 0.46% in fRe as well as 0.53% in overall Nusselt number. Fig. 3 demonstrates that the Nusselt number distributions along streamwise direction are almost overlapped in the cases of the two finest grids. That means the grid independence is established for the case with grid size $30 \times 30 \times 390$. Therefore, the moderate grid size ($30 \times 30 \times 390$) is chosen in the present work.

4. Results and discussion

4.1. Numerical validation

To assess the accuracy of the current numerical model, it is essential

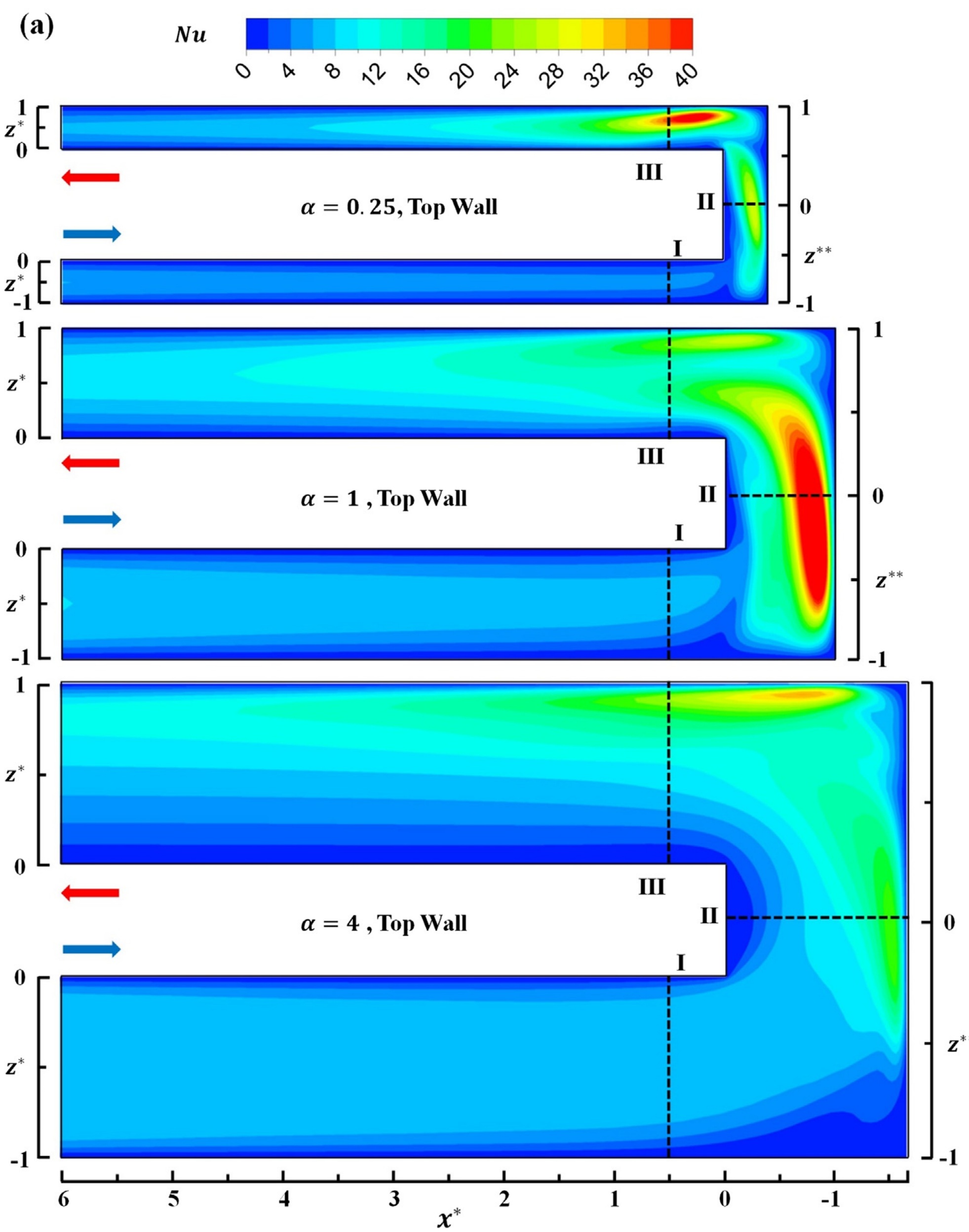

Fig. 5. (a) Nu contours on top and bottom walls, (b) corresponding near-wall flow fields, and (c) secondary flow vectors and dimensionless temperature (T^*) contours at three selected stations (I, II, and III) for $\alpha = 0.25$, 1, and 4 at $\theta = 90^{\circ}$ and $Re = 200$.

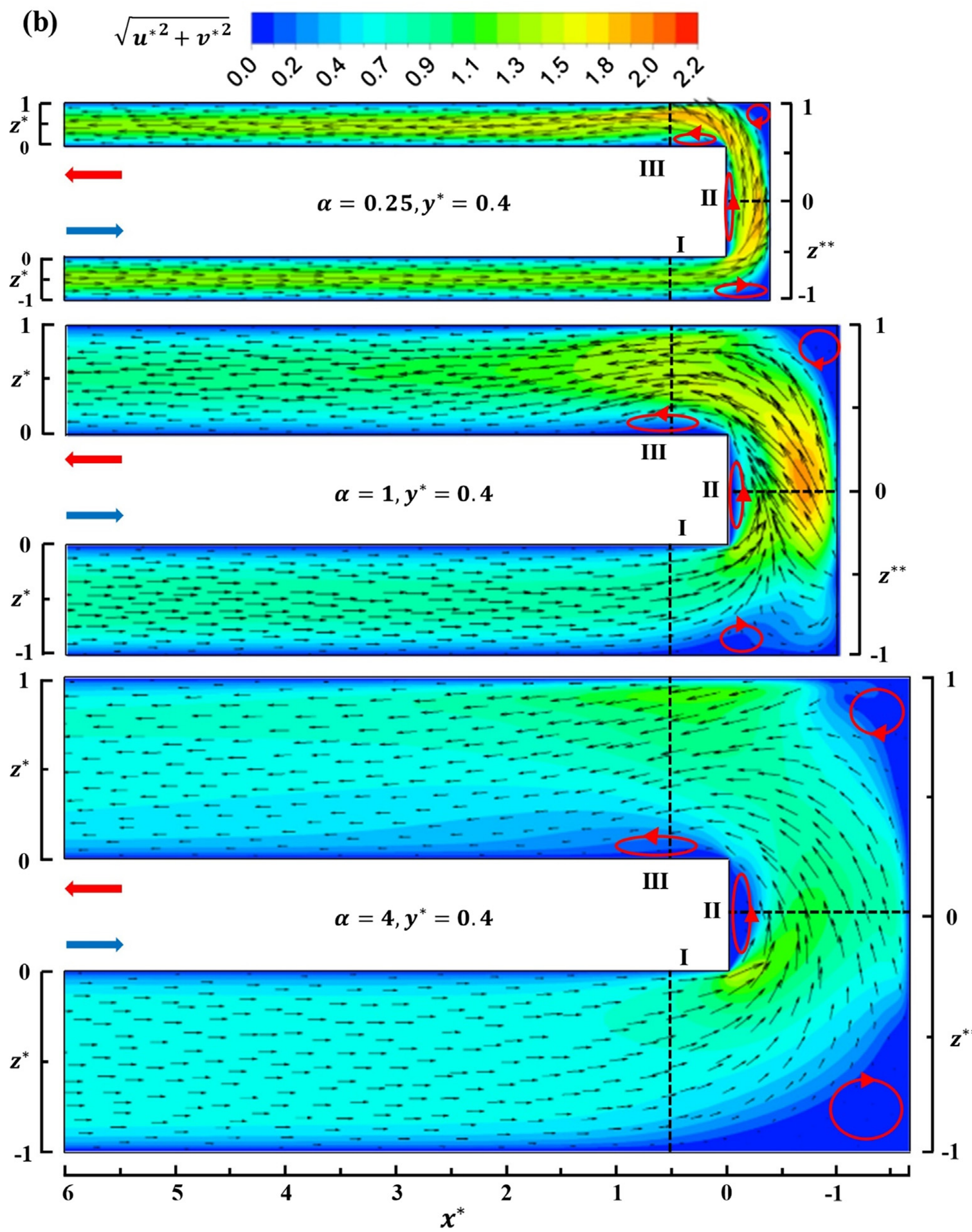

Fig. 5. (*continued*)

to validate the present results with available data. The fully developed straight rectangular duct flows with α ranging from 1 to 10 are validated under constant wall temperature boundary condition with the data published by Schmidt and Newell [35], Shah [36], and Aparecido and Cotta [37]. Fig. 4 shows that the predicted values of *Nu* are in good agreement with those reported by previous researchers.

4.2. Effect of aspect ratio

Fig. 5(a)–(c) respectively plot the local *Nu* contours on top walls, near-wall velocity vectors on $y^* = 0.4$ planes, and secondary flow patterns and dimensionless temperature (T^*) at station I ($x^* = 0.5$, $z^{**} < 0$), II ($z^{**} = 0$), and III ($x^* = 0.5$, $z^{**} > 0$) for $\alpha = 0.25$, 1, and 4 under $\theta = 90^\circ$ and $Re = 200$. They are chosen because the other cases show qualitatively similar trend of hydrothermal variations with respect to θ. In addition, only the results on or near top walls are presented due to the symmetry of the 90° channel.

At the entrance of the first pass ($x^* = 6$), the near-developed flow is traveling downstream and continuously heated by the top walls as depicted in Fig. 5(a) and (b). The channel shows highest *Nu* values (pseudo cyan color) in the center ($z^* = -0.5$) and lowest ones near outer wall ($z^* = -1$) and inner wall ($z^* = 0$) for all channels (Fig. 5(a)) due to the effect of corner (right angle) where the deep blue color suggests noticeable poor heat transfer (low *Nu*). With the development of thermal boundary layer, the low *Nu* region grows. Near the entrance of 180° sharp turn (or around station I) the size of the low *Nu* region (around $z^{**} = -1$) on top wall of $\alpha = 4$ channel is apparently smaller than those of $\alpha = 1$ and 0.25 channels. This is because the turning of fluid flow into the 180° sharp bend forces the fluids to move away from the outer wall and the larger α tend to aid the smooth turning of fluid flow thus resulting in higher heat transfer, as evidenced by Fig. 5(b). Relative to $\alpha = 4$ case, the turning of fluid flow into the sharp bend tends to force the fluids to travel toward the inner wall of $\alpha = 1$ and 0.25 channels, and, in turn, brings the corner vortex back to the first pass, enlarging the area of poor heat transfer region. To reveal the turn effect, the near top wall vectors and velocity contours plotted in

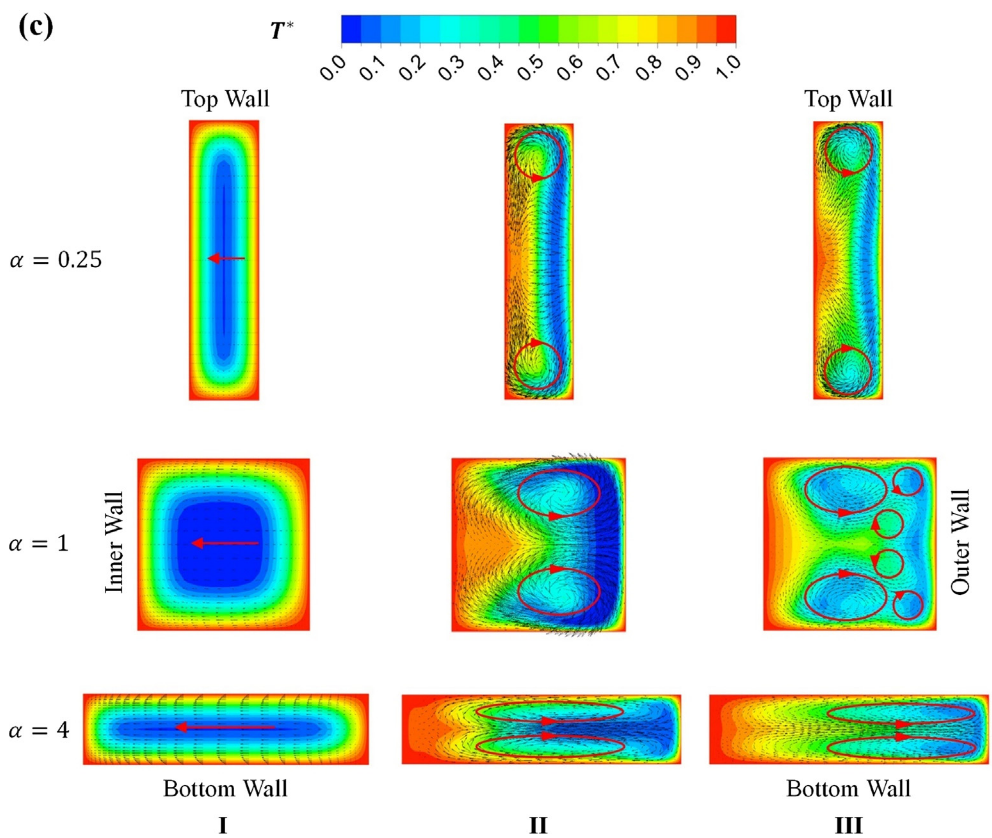

Fig. 5. (*continued*)

Fig. 5(b) clearly demonstrate the low-velocity regions with the corner vortex, in a decent match with low *Nu* regions indicated in Fig. 5(a). The turn effect can also be observed in the form of temperature contours and secondary flow vectors at the station I ($x^* = 0.5$, $z^* = -1$ to 0) as displayed in the first column of Fig. 5(c). It is noticeable that the four right corners are filled with high-temperature low-speed fluids. Moreover, the secondary flow velocity magnitude for $\alpha = 4$ channel is much greater than those of $\alpha = 0.25$ and 1 channels, which means that the turn effect for $\alpha = 4$ case is more prominent.

With the high-speed fluid entering the 180° sharp bend from the first pass in Fig. 5(b), there exist two acceleration regions, one near the channel tip ($z^{**} = 0$, hereinafter called first acceleration region) and the other near the outer wall of second pass ($z^{**} = 1$, hereinafter called second acceleration region). The two acceleration regions produce two distinctly high *Nu* areas for all three α presented (Fig. 5(a)). For $\alpha = 1$ and 0.25 channels, the first acceleration region (Fig. 5(b)) flushes the near top wall corner vortex from the front-turning corner ($x^* = -1$ and -0.5, $z^{**} = -1$) back to the first pass at ($x^* = 0$, $z^{**} = -0.9$) resulting in poor heat transfer in that region (Fig. 5(a)). However, the mild acceleration in the $\alpha = 4$ (Fig. 5(b)) results in a much larger low *Nu* region on top wall than those in the $\alpha = 1$ and 0.25 cases (Fig. 5(a)). The relative high *Nu* around the mid-turn (station II) of $\alpha = 1$ and 0.25 channels (Fig. 5(a)) is ascribed to downwash cooling as depicted by the secondary flow vectors in Fig. 5(c). Compared to the two weak counter-rotating vortices (Dean vortices) in the $\alpha = 4$ channel, the Dean vortices in $\alpha = 1$ and 0.25 channels are much stronger. The strong vortex pair which mixes the fluid well is attributable to higher *Nu* enhancement than that in $\alpha = 4$ case. For all cases, there exists a low *Nu* region near the divider tip of the top wall (Fig. 5(a)) because of the separation bubble near the tip in Fig. 5(b). In the rear half of the turning region ($z^{**} = 1$), there is a rear turning corner vertex around $z^{**} = 1$ in all cases presented as shown in Fig. 5(b). The existence of rear turning corner vertex suppresses heat transfer in that area as depicted in Fig. 5(a). Meanwhile, the $\alpha = 1$ channel provides the highest *Nu* augmentation in the turn region, compared with the $\alpha = 4$ and 0.25 channels (the relative *Nu* levitation in the bend of $\alpha = 1$ channel is 43% higher than that of $\alpha = 4$ channel). This is due the fact that the near-top wall convective flow field (Fig. 5(b)) and second flow (Fig. 5(c)) in the mid-turn of $\alpha = 1$ channel are much stronger in terms of velocity vectors or contour levels.

In the second pass, there exists a small separation bubble (Fig. 5(b)) at station III for all cases. The occurrence of the separation bubble leads to a poor heat transfer region near the inner wall ($x^* = 0.5$, $z^* = 0$). With the fluid flow moving downstream, the strength of Dean vortices reduces, and again the heat transfer performance is dominated by convection and cross-sectional shape, i.e. high *Nu* around the center and low *Nu* adjacent to the walls.

4.3. Effect of included angle

In order to illustrate the effect of θ, the local *Nu* contours on top and bottom walls, near-wall velocity vectors on $y^* = \pm 0.4$ planes, and secondary flow patterns and dimensionless temperature (T^*) at station I, II, and III for $\theta = 45$ and 75° under $\alpha = 1$ and $Re = 200$, are respectively depicted in Fig. 6(a)–(c). Similar to rectangular channels in Fig. 5, the near-developed flow in the first pass of parallelogram channels moves downstream and continues to be heated by top and bottom walls. Consequently, there is a relative high heat transfer area (pseudo cyan color) around the top and bottom wall centers at the entrance ($x^* = 6$). It is noticeable that the cyan color on top and bottom walls of $\theta = 45°$ channel is asymmetric and does not respectively extend to inner wall ($z^* = 0$) or outer wall ($z^* = -1$) (i.e. poor heat transfer near the acute angle sides) since the flow is restrained near the acute angle sides. When the thermal and velocity boundary layers grow, the afore-mentioned low *Nu* region persists and increases. Nonetheless, the low *Nu* area ($z^{**} = -1$) near the entrance of 180° turn of the

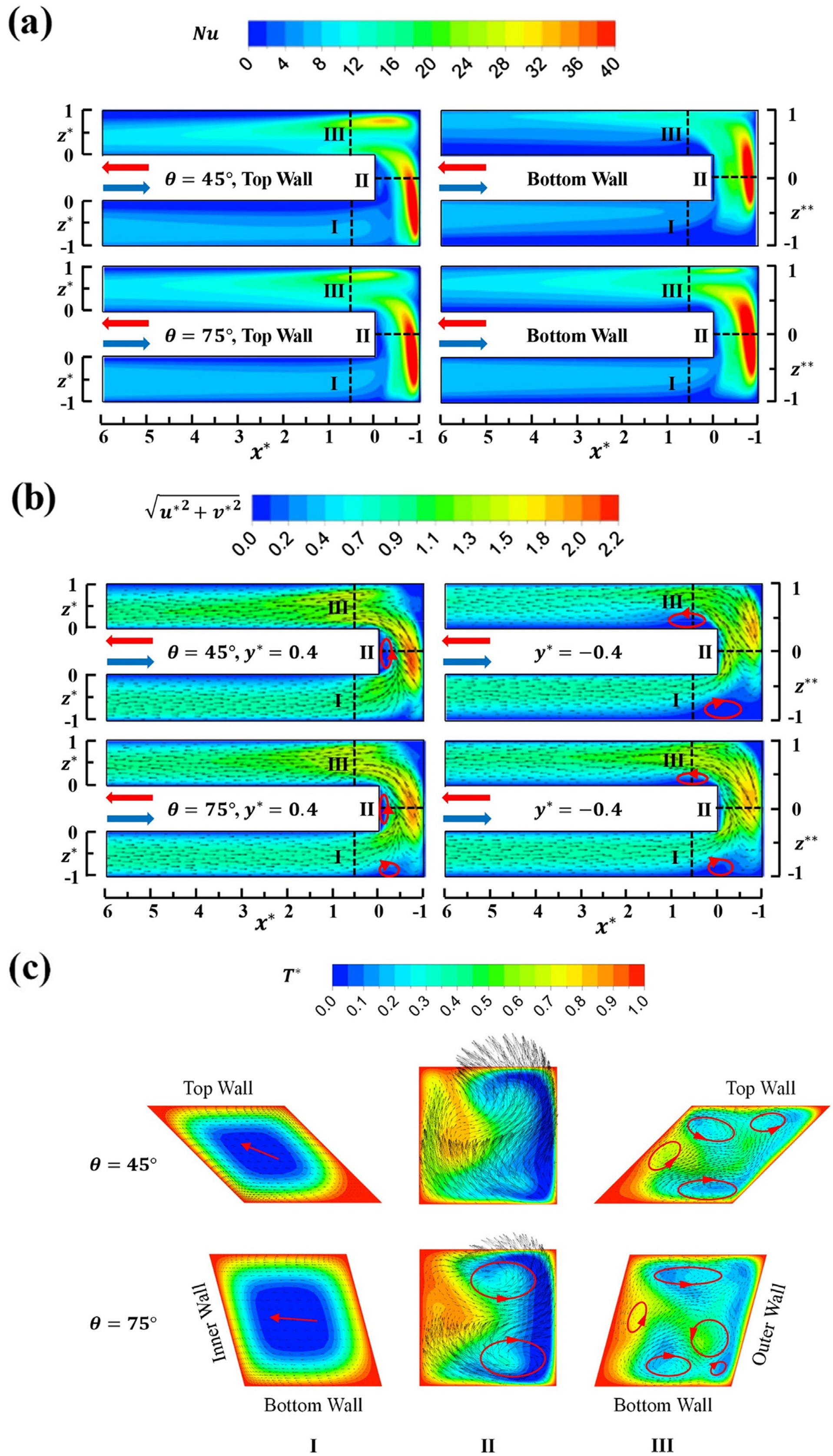

Fig. 6. (a) *Nu* contours on top and bottom walls, (b) corresponding near-wall flow fields, and (c) secondary flow vectors and dimensionless temperature (T^*) contours at three selected stations (I, II, and III) for $\theta = 45^\circ$ and 75° at $\alpha = 1$ and $Re = 200$.

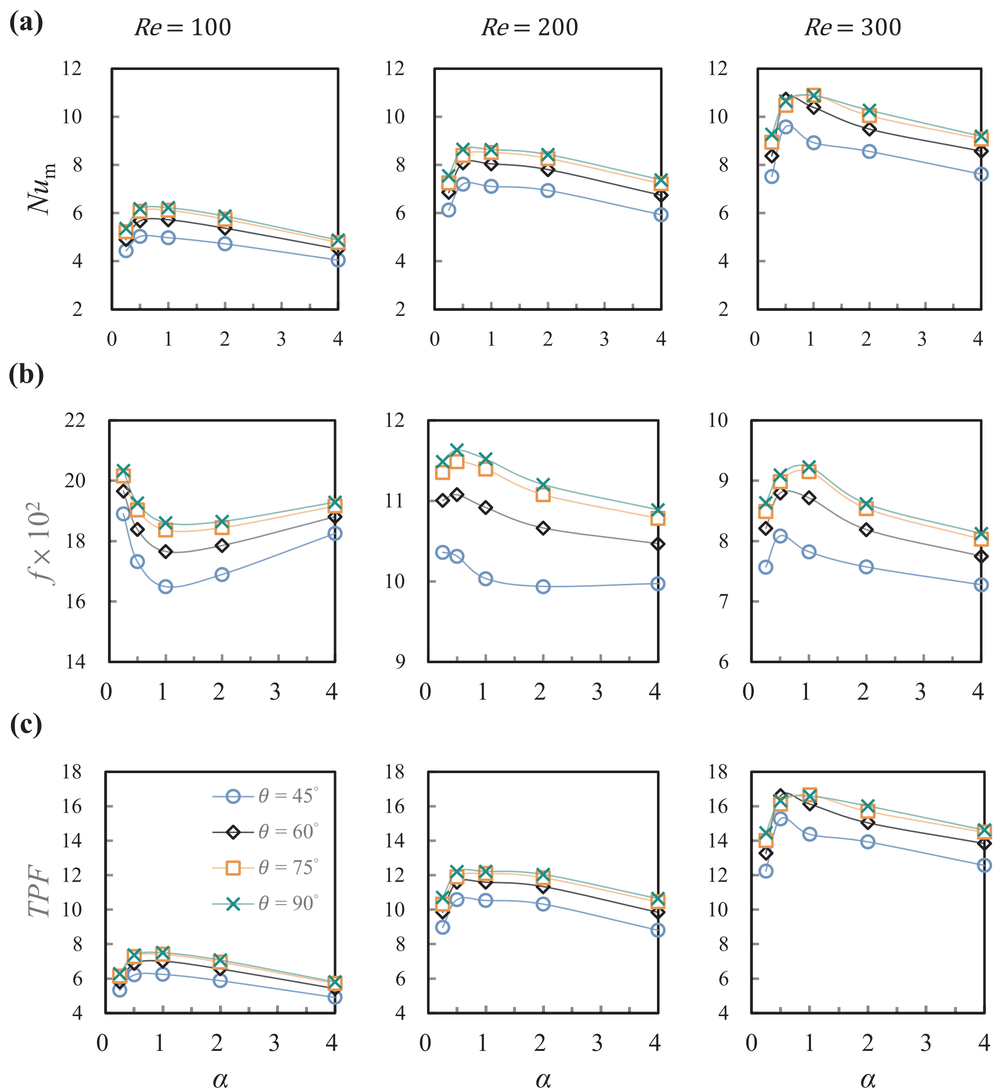

Fig. 7. Numerical predictions of (a) Nu_m, (b) f, and (c) TPF for parallelogram two-pass channels with various θ, α, and Re.

Table 3
Average Nusselt number (Nu_m) correlation coefficients.

i	Coefficients of $Nu_m Re$ correlation, $C_{i,j}$				
	$j = 0$	1	2	3	4
0	−0.0222	0.9232	−1.143	0.494	−0.06598
1	0.008085	0.001938	-4.297×10^{-4}	6.211×10^{-6}	
2	-4.486×10^{-5}	-1.195×10^{-5}	2.503×10^{-6}		

bottom wall in 45° channel (Fig. 6(a)) is apparently much larger than that of 90° channel (Fig. 5(a)). This means that the turning of fluid flow into the 180° sharp bend augments the effect of acute angle on *Nu* deterioration since it tends to force the fluids to move away from the outer wall of 45° channel. In comparison to the bottom wall, the turning of fluid flow into the sharp bend near the top wall of the 45° channel tends to force the fluids to move toward the outer wall, thus counteracting the impact of corner vortex on *Nu* deterioration ($z^{**} = -1$) and reducing the poor heat transfer area (Fig. 6(a)). The 75° channel demonstrates similar hydrothermal trends, except that the acute angle effect is weaker than that of the 45° channel.

The above heat transfer distributions can be explained by the near-wall velocity vectors and velocity contours plotted in Fig. 6(b). It is clearly seen that the low-velocity regions near the acute angle side associate with low *Nu* regions as illustrated in Fig. 6(a). The acute angle effect can also be observed by inspecting the secondary flow field vectors and temperature contours at station I in Fig. 6(c). The acute corners between the outer wall and bottom wall are found to be occupied with high-temperature low-speed fluids. Moreover, the high-temperature region near the acute-angle for $\theta = 45°$ is much larger than that for $\theta = 90°$ (Fig. 5(c)). Compared with $\theta = 45°$ and 90° channels, the turn effect in 75° channel reduces the acute angle effect near bottom wall and leads to less *Nu* deterioration.

Similar to 90° channel, the high-speed fluids are further accelerated near the tip ($z^{**} = 0$, hereinafter called first acceleration) and outer wall of second pass ($z^{**} = 1$, hereinafter called second acceleration) as they turn from the first pass into the bend in Fig. 6(b). The two consecutive accelerations produce two distinctly high *Nu* areas on the bottom wall for all θ presented (Fig. 6(a)). Focusing on 45° channel, the near-bottom-wall corner vortex cannot be completely washed away from the front turning corner ($x^* = -1$, $z^{**} = -1$) back to the first pass at ($x^* = 0$, $z^{**} = -0.9$) by the first acceleration (Fig. 6(b)), resulting in poor heat transfer in those regions (Fig. 6(a)). This is due to the further location of the first acceleration away from the front corner. In contrast, the first acceleration location in the 90° case (Fig. 5(b)) is closer to the front corner, resulting in relatively smaller corner vortex and, in turn, a slightly higher *Nu* distribution ((Fig. 5(a))) than that in the 45° case as indicated in Fig. 6(a). Nevertheless, the acute angle in

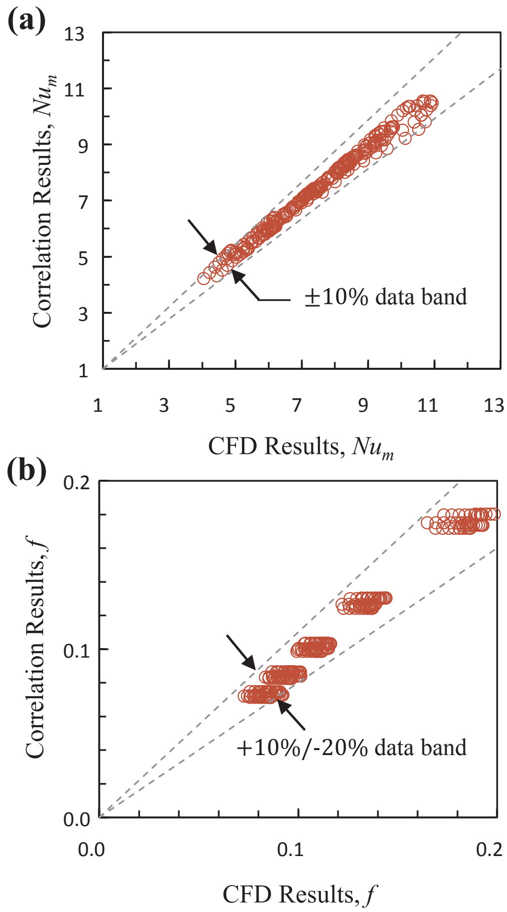

Fig. 8. Comparison of the CFD predicted and correlated (a) Nu_m and (b) f.

Table 4
Friction factor (f) correlation coefficients.

i	Coefficients of fRe correlation, $C_{i,j}$				
	$j = 0$	1	2	3	4
0	4.983	−1.107	−0.593	0.5897	−0.09281
1	0.06307	0.04947	−0.01964	1.684×10^{-3}	
2	-3.727×10^{-4}	-2.202×10^{-4}	5.969×10^{-5}		

the front turn of 75° channel, has less influence on the location and size of the front turning corner vortex near the bottom wall, which induces less Nu deterioration in that region.

Around the station II or mid-turn, there exists a relative high Nu region on the bottom wall of 45° channel (Fig. 6(a)) due to the impingement cooling depicted by the secondary flow vectors in Fig. 6(c). Relative to a pair of counter-rotating Dean vortices in 90° channel, no indication of vortex can be found in 45° channel. The lack of vortex pair is responsible for less Nu enhancement than that in 90° channel. As for 75° channel, the Dean vortex pair is very similar to that of the 90° channel except the slight asymmetry. In addition, there is a low Nu region near the divider tip of the top wall (Fig. 6(a)) resulting from the corresponding separation bubble in Fig. 6(b). With the second acceleration of flow near the outer wall ($z^{**} = 1$), no rear turning corner vertex near the top walls of 45° and 75°channels is observed (Fig. 6(b)). Consequently, the Nu in that region is augmented as depicted in Fig. 6(a). Overall, the heat transfer performance in the turn region of 90° channel is superior to those of 45 and 75° cases since the Dean vortices in 90° case are more intact (Figs. 5(c) and 6(c)).

Unlike the 90° and 75° channels, the weak downwash ($x^* = 0$, $z^{**} = 0.9$) after the second acceleration is absent near the bottom wall of 45° channel as one can observe from the secondary flow velocity vector (Fig. 6(c)) at station III. Due to the absence of downwash on the bottom wall, the heat transfer distribution (Fig. 6(a)) in the second pass of 45° channel are much lower than those in 75° and 90° channels. Moreover, the combined effects of acute angle and separation bubble near the bottom wall (Fig. 6(b)) deteriorate heat transfer near the inner wall ($x^* = 0.5$, $z^* = 0.1$). With decaying turning effect in the second pass, the convection and acute angle effect again determine the heat transfer distribution similar to the first pass.

4.4. Nusselt number and friction factor correlations

To identify the optimal α and θ with best heat transfer, the overall Nusselt number (Nu_m) are evaluated by weighting the local Nu against the bottom and top walls for $Re = 100$, 200, and 300 as depicted in Fig. 7(a). For all Re examined, it is seen that the $\alpha = 0.5$ case gives the highest value of Nu_m due to the strong acceleration and Dean vortices mentioned in Section 4.2. As α increases from 0.25 to 4 for all included angles, the heat transfer increases with α for $\alpha < 0.5$ and then decreases for $\alpha > 0.5$. In addition, the heat transfer increases with increasing Re. In terms of the included angle, the heat transfer augmentation of 90° case is superior to any other cases because of smaller hydraulic diameter and weaker acute angle effect. The maximum Nu_m augmentation of 90° channel is 23% higher than that of 45° channel. A combined polynomial correlation of Nu_m with respect to α, θ, and Re are formulated in Eq. (19) with corresponding coefficients listed in Table 3. Fig. 8(a) illustrates the comparison of simulated and correlated Nu_m. It is noticeable that the correlation results are in good agreements with the CFD data, with a maximum error below 10% (see Table 4).

$$Nu_{\mathrm{m}} = Re^{0.5} \sum_{j=0}^{5} \sum_{i=0}^{5} C_{i,j} \alpha^i \theta^j \tag{19}$$

Fig. 7(b) shows the Fanning friction factor f against α, θ for all Re examined. It is seen that the 45° case shows the lowest value of f since the division of dimensionless distance eliminates the effect of D_h. As Re increases from 200 to 300, the trend of f with α is reversed. For $Re = 100$, the lowest value of f occurs at $\alpha = 1$ and for $Re > 200$, the minimal location is changed. Hence, there exists a critical value Re_c around $Re = 200$ for f. Compared with that of 90° channel, the maximum f reduction of 45° channel is 18%. Similar to Nu_m, a combined polynomial correlation of f with respect to α, θ, and Re are also formulated in Eq. (20). The corresponding coefficients are listed in Table 3. The maximum error of the correlation compared to the CFD results is within 20% (Fig. 8(b)), which shows reasonable agreement.

$$f = Re^{-0.8} \sum_{j=0}^{5} \sum_{i=0}^{5} C_{i,j} \alpha^i \theta^j \tag{20}$$

Since the above coefficients are calculated under constant flow rate condition, the thermal performance factor evaluated under constant pumping power consumption, i.e. TPF, should also be considered. Fig. 7(c) displays the TPF profiles with different α, θ, and Re. It is noticeable that the maximum value of TPF appears at $\alpha = 0.5$ and $\theta = 90°$ for all cases examined due to the best heat transfer performance and relatively lower f of 90° channel. In contrast, the 45° channel shows the lowest TPF values. Additionally, all TPF values are augmented as Re increases from 100 to 300.

5. Conclusions

The present numerical approach has been well validated by comparing its results with the analytical and CFD data available from the literature for $\theta = 90°$ and $1 \le \alpha \le 10$ under the constant wall temperature condition. The maximum relative error is within 5%. Within the range of α, θ, and Re examined, some new findings are drawn based on the predicted laminar heat and fluid flow features.

1. In the first pass of the channel, heat transfer increases with increasing values of α. Nonetheless, in the turn and second pass, higher Nu augmentations are observed for channels with α around unity due to the stronger Dean vortex pair induced by the 180° sharp turn. The relative Nu levitation in the bend of $\alpha = 1$ channel is 43% higher than that of $\alpha = 4$ channel.
2. The Nu distributions on bottom wall and top wall in the slant wall channels ($\theta \neq 90°$) are unequal and lower than that of 90° channel (rectangular channel) due to the acute angle effect, a new finding not reported previously.
3. For fixed θ and Re, Nu_m increases and then decreases with increasing values of α. The peak Nu_m is located around $\alpha = 0.5$. Furthermore, Nu_m increases monotonically with increasing values of θ for a constant α and Re. The 90° channel demonstrates the highest overall heat transfer performance with a maximum Nu_m augmentation of 23% for all α, θ, and Re examined. Similar trends are also found in TPF.
4. There exists a critical Reynolds number, Re_c around $Re = 200$. Below Re_c, the trend of f against α is reversed in comparison with that of Nu_m, i.e. decreasing and then increasing with increasing values of α. However, beyond Re_c the profiles of f are in accordance with those of Nu_m. In addition, the variation of f with respect to θ is parallel with that of the Nu_m, which results in a maximum friction reduction of 18%.
5. New correlations of Nu and f with respect to α, θ, and Re are proposed in the present study for laminar flow in parallelogram serpentine micro heat exchangers. The maximum errors of the correlations for Nu and f compared to the CFD results are within 10% and 20%, respectively.

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References

[1] S.G. Kandlikar, Z. Lu, Thermal management issues in a PEMFC stack – a brief review of current status, Appl. Therm. Eng. 29 (7) (2009) 1276–1280.
[2] A. Faghri, Z. Guo, Challenges and opportunities of thermal management issues related to fuel cell technology and modeling, Int. J. Heat Mass Transf. 48 (19–20) (2005) 3891–3920.
[3] F.C. Chen, Z. Gao, R.O. Loutfy, M. Hecht, Analysis of optimal heat transfer in a PEM fuel cell cooling plate, Fuel Cells 3 (4) (2003) 181–188.
[4] G. Inoue, T. Yoshimoto, Y. Matsukuma, M. Minemoto, H. Itoh, S. Tsurumaki, Effect of flow pattern of gas and cooling water on relative humidity distribution in polymer electrolyte fuel cell, J. Power Sources 162 (1) (2006) 94–104.
[5] C. Xu, T.S. Zhao, A new flow field design for polymer electrolyte-based fuel cells, Electrochem. Commun. 9 (3) (2007) 497–503.
[6] S.H. Yu, S. Sohn, J.H. Nam, C.J. Kim, Numerical study to examine the performance of multi-pass serpentine flow-fields for cooling plates in polymer electrolyte membrane fuel cells, J. Power Sources 194 (2) (2009) 697–703.
[7] J.C. Kurnia, A.P. Sasmito, A.S. Mujumdar, Numerical investigation of laminar heat transfer performance of various cooling channel designs, Appl. Therm. Eng. 31 (6–7) (2011) 1293–1304.
[8] W.R. Dean, Note on the motion of fluid in a curved pipe, The London, Edinburgh, Dublin Philos. Mag. J. Sci. 4 (20) (1927) 208–223.
[9] S. Dennis, N.G. Michael, Dual solutions for steady laminar flow through a curved tube, Quart. J. Mech. Appl. Math. 35 (3) (1982) 305–324.
[10] L. Wang, F. Liu, Forced convection in slightly curved microchannels, Int. J. Heat Mass Transf. 50 (5–6) (2007) 881–896.
[11] T.-S. Wang, M.K. Chyu, Heat convection in a 180-deg turning duct with different turn configurations, J. Thermophys. Heat Transfer 8 (3) (1994) 595–601.
[12] S. Chintada, K.-H. Ko, N.K. Anand, Heat transfer in 3-D serpentine channels with right-angle turns, Numer. Heat Transfer, Part A: Appl. 36 (8) (1999) 781–806.
[13] N.R. Rosaguti, D.F. Fletcher, B.S. Haynes, Laminar flow and heat transfer in a periodic serpentine channel, Chem. Eng. Technol. 28 (3) (2005) 353–361.
[14] Y. Lasbet, B. Auvity, C. Castelain, H. Peerhossaini, A chaotic heat-exchanger for PEMFC cooling applications, J. Power Sources 156 (1) (2006) 114–118.
[15] H. Aref, Stirring by chaotic advection, J. Fluid Mech. 143 (1984) 1–21.
[16] Z. Zheng, D.F. Fletcher, B.S. Haynes, Chaotic advection in steady laminar heat transfer simulations: periodic zigzag channels with square cross-sections, Int. J. Heat Mass Transf. 57 (1) (2013) 274–284.
[17] Z. Zheng, D.F. Fletcher, B.S. Haynes, Laminar heat transfer simulations for periodic zigzag semicircular channels: Chaotic advection and geometric effects, Int. J. Heat Mass Transf. 62 (2013) 391–401.
[18] Z. Zheng, D.F. Fletcher, B.S. Haynes, Transient laminar heat transfer simulations in periodic zigzag channels, Int. J. Heat Mass Transf. 71 (2014) 758–768.
[19] P.E. Geyer, D.F. Fletcher, B.S. Haynes, Laminar flow and heat transfer in a periodic trapezoidal channel with semi-circular cross-section, Int. J. Heat Mass Transf. 50 (17–18) (2007) 3471–3480.
[20] R. Gupta, P.E. Geyer, D.F. Fletcher, B.S. Haynes, Thermohydraulic performance of a periodic trapezoidal channel with a triangular cross-section, Int. J. Heat Mass Transf. 51 (11–12) (2008) 2925–2929.
[21] N.R. Rosaguti, D.F. Fletcher, B.S. Haynes, Low-Reynolds number heat transfer enhancement in sinusoidal channels, Chem. Eng. Sci. 62 (3) (2007) 694–702.
[22] A.M. Guzmán, M.J. Cárdenas, F.A. Urzúa, P.E. Araya, Heat transfer enhancement by flow bifurcations in asymmetric wavy wall channels, Int. J. Heat Mass Transf. 52 (15–16) (2009) 3778–3789.
[23] R.K. Shah, A.L. London, Laminar Flow Forced Convection in Ducts, Academic Press, New York, 1978.
[24] N.R. Rosaguti, D.F. Fletcher, B.S. Haynes, Laminar flow and heat transfer in a periodic serpentine channel with semi-circular cross-section, Int. J. Heat Mass Transf. 49 (17–18) (2006) 2912–2923.
[25] H. Ju, C.-Y. Wang, S. Cleghorn, U. Beuscher, Nonisothermal modeling of polymer electrolyte fuel cells: I. experimental validation, J. Electrochem. Soc. 152 (8) (2005) A1645–A1653.
[26] D. Cvetković, M. Bojić, Optimization of thermal insulation of a house heated by using radiant panels, Energy Build. 85 (2014) 329–336.
[27] T.-M. Liou, S.-W. Chang, S.-P. Chan, Y.-S. Liu, K.-C. Chu, Flow visualization and PIV measurements in a two-pass smooth-wall parallelogram channel, The 9th Pacific Symposium on Flow Visualization and Image Processing, Busan, Korea, (2013).
[28] T.-M. Liou, C.-C. Chen, M.-Y. Chen, TLCT and LDV measurements of heat transfer and fluid flow in a rotating sharp turning duct, Int. J. Heat Mass Transf. 44 (9) (2001) 1777–1787.
[29] T.-M. Liou, S.-W. Chang, C.-C. Yang, Y.-A. Lan, Thermal performance of a radially rotating twin-pass smooth-walled parallelogram channel, J. Turbomach. 136 (12) (2014) 121007.
[30] T.-M. Liou, C.-S. Wang, S.-P. Chan, Effect of included angle on turbulent flow and heat transfer in rhombic serpentine heat exchangers, Int. J. Therm. Sci. 114 (2017) 155–171.
[31] T.-M. Liou, H. Wang, S.-P. Chan, Nusselt number as composite functions of aspect ratio and wall inclination in parallelogram microchannels with three types of thermal boundary conditions, J. Mech. 33 (4) (2017) 521–533.
[32] ANSYS Inc., ANSYS Fluent User's Guide, 2014.
[33] B.P. Leonard, A stable and accurate convective modelling procedure based on quadratic upstream interpolation, Comput. Methods Appl. Mech. Eng. 19 (1) (1979) 59–98.
[34] V.V. Dharaiya, S.G. Kandlikar, Numerical investigation of heat transfer in rectangular microchannels under H2 boundary condition during developing and fully developed laminar flow, J. Heat Transf. 134 (2) (2011) 020911-020911.
[35] F.W. Schmidt, M.E. Newell, Heat transfer in fully developed laminar flow through rectangular and isosceles triangular ducts, Int. J. Heat Mass Transf. 10 (8) (1967) 1121–1123.
[36] R.K. Shah, Laminar flow friction and forced convection heat transfer in ducts of arbitrary geometry, Int. J. Heat Mass Transf. 18 (7–8) (1975) 849–862.
[37] J.B. Aparecido, R.M. Cotta, Thermally developing laminar flow inside rectangular ducts, Int. J. Heat Mass Transf. 33 (2) (1990) 341–347.